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标题	Palladium-Catalyzed Carbonylative Sonogashira Coupling of Aliphatic Alkynes and Aryl Thianthrenium Salts to Alkynones and Furanones
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摘要（原文节选）

Herein, we developed a mild and efficient palladium-catalyzed carbonylative Sonogashira coupling of aryl thianthrenium salts with aliphatic alkynes and benzyl acetylene toward alkynones and furanones. Various desired products were prepared in good yields with broad functional group tolerance including the bromide group. In the case of using benzyl acetylene, the corresponding furanones can be obtained in good yields under the same conditions with two molecules of CO inserted. ⋯

数据表 · Table 1（19 行，含数值行 19 行）

Optimization of the Reaction Conditionsa

Entry	Variation from the Standard conditions	Yield (%)b
1	None	78 (72)c
2	No ligand	n.o.
3	No catalyst	n.o.
4	No base	n.o.
5	DABCO instead DIPEA	56
6	Cs2CO3 instead of DIPEA	n.o.
7	K3PO4 instead of DIPEA	n.o.
8	Pd(OAc)2 instead of PdI2	65
9	PdBr2 instead of PdI2	65
10	(cinnamyl)2Pd2Cl2 instead of PdI2	61
11	PCy3 instead of PPh3	39
12	BuPAd2 instead of PPh3	63
13	BINAP instead of PPh3	trace
14	DCE as solvent	61
15	DMSO as solvent	62
16	CH3CN as solvent	40
17	0.2 mol L − 1 DMF	66
18	1.0 equiv. 2a	70
19	CuI (10 mol %) as additive	trace

脚注（逐字）：aThe reaction was conducted using 1a (0.2 mmol), 2a (x mmol), catalyst (2 mol %), ligand (6 mol %), base (2.0 equiv), CO (10 bar), 80 ° C, 15 h.bDetermined by GC with hexadecane as internal standard.cIsolated yield is shown in parentheses.

正文对应句（逐字）：Initially, we started with readily available aryl thianthrenium salt 2a and heptane 1a as the model substrates to establish this reaction, and the target product 3a was obtained (Table 1). After a systematic investigation of the reaction parameters, we determined the optimal conditions: alkyl alkyne

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